

GENERAL SPECIFICATION G4.60

Pressure Vessels

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G4.60 PRESSURE VESSELS

G4.60.1 General

- (a) This specification defines the mandatory technical requirements for pressure vessels.
- (b) Performance and design requirements covered in Pressure Vessel Data Sheet(s), if supplemented by the S.O., shall also be complied with. In the event of any conflicting requirement, these shall be brought to the notice of the S.O. in writing. When the S.O. requests the Contractor/Manufacturer to furnish a vessel conforming to the Manufacturer's proprietary design, the materials of construction and other construction details shall be selected by the Manufacturer based on the design conditions specified by the Board or the S.O..
- (c) Unless otherwise stated, the manufacturer's drawings and calculations shall use S.I. units of measurement.
- (d) Internal and external design pressures shall be according to the specific applicable range of process operating requirements for which the vessel is used with incorporation of allowance factors according to vessel design codes. For internal pressure, static liquid head from the highest operating liquid level shall be added.
- (e) Maximum and minimum design temperatures shall be as specified according to the applicable process requirements with the necessary allowance factors governed by vessel design codes. Minimum Metal Design Temperature (MDMT); when not specified, shall be based on the Lowest One Day Mean Ambient Temperature where the vessel is located or coldest operating temperature, whichever governs.

G4.60.2 Codes and Standards

- (a) It shall be noted that where codes, standards, their articles, diagrams, figures, tables, etc. are referenced, the onus is upon the Contractor to ensure that the latest updated edition is being used in performing all the obligations relating to this part of the Contract relating to pressure vessels.
- (b) The documents that are listed below and referenced herein apply with regard to the design, supply and installation of pressure vessels for the Contract. Codes and industry standards are listed under their responsible Society and, if prefixed by "ANSI," are American National Standards":
 - (i) American Society of Mechanical Engineers (ASME)
 - 1) ANSI/ASME Boiler and Pressure Vessel Code, Current Edition, including all mandatory addenda in effect on the date of Purchase Order:

Section II	"Material Specifications"
Part A	Ferrous Materials
Part B	Non-Ferrous Materials
Part C	Welding Rods, Electrodes and Filler Materials
Part D	Properties

Section VIII	"Pressure Vessels"
Division 1	Requirements applicable to the design, fabrication, inspection, testing, and certification of pressure vessels operating at either internal or external pressures. Requirements on various classes of material used in pressure vessel construction and for fabrication methods such as welding, forging and brazing. Rules pertaining to the use of the single ASME certification mark with the U, UM and UV designators
Division 2	Alternative rules

- (ii) BS EN 1092-1 Flanges and their joints – Circular flanges for pipes, fittings and accessories, PN designed – Part 1: Steel flanges.

G4.60.3 Design Conditions

Vessels and their supports shall be designed to resist the effects of the following combinations of loads within the limits of allowable stress specified herein. Wind or dynamic loads shall not be assumed to occur simultaneously:

- (a) Empty Condition: The empty vessel weight plus the greater of wind or seismic load.
- (b) Operating Condition: The operating weight of the vessel at design pressure plus any attached items (piping, valves, heat exchangers, etc.), plus the greater of wind or seismic loads.
- (c) Field Hydrotest Condition: The test pressure, the weight of the vessel in the fully corroded condition, attached equipment, and the weight of the testing contents in the vessel and any other equipment as may be attached, plus 1/3 of the design wind load.

G4.60.4 Design Allowable Stresses

For the design conditions given, the calculated stresses shall be maintained within the following limits:

- (a) Operating (the design metal temperature with maximum operating liquid and internal loading): The basic allowable stresses permitted by the applicable Code.
- (b) Test (based on the nominal thickness at the test temperature): For pressure parts, 90 % of specified minimum yield strength, except that austenitic stainless steels may be stressed up to 95 % of specified minimum yield strength; for non-pressure parts, the basic Code allowable stresses increased by 33-1/3 %.
- (c) Empty erected vessel: The basic allowable stresses increased by 20 %.
- (d) All design conditions: The maximum allowable longitudinal compressive stress for support skirts shall be determined in accordance with ASME Section VIII, Division 1, Paragraph UG-23(b).

- (e) Anchor bolt stress: Carbon steel anchor bolting shall be designed for an allowable stress based on the corroded root area of the thread. The allowable stress for the anchor bolts shall be as follows:

Table G4.60-1: Anchor Bolt Allowable Stresses

For vessels with a vertical height $H < 14$ m	137,896 kN/m ²
For vessels with a vertical height $14 \text{ m} < H < 31$ m	124,106 kN/m ²
For vessels with a vertical height $H > 31$ m; Vessels in sour service; Vessels with a corrosion allowance > 3 mm; Vessels in severe corrosion environments	103,422 kN/m ²
Horizontal vessels	124,106 kN/m ²

- (f) Welds attaching non-pressure parts to pressure parts: Welds attaching non-pressure parts to the pressure shells and supports for internal equipment such as cyclones or grids, etc. shall be designed to the allowable stresses for pressure parts.

G4.60.5 Materials of Construction

- (a) All materials of construction shall be new and as specified.
- (b) Vessel materials will normally be specified on the vessel data sheets which may not necessary be available at the same time as this specification. Substitution of materials for any part shall not be allowed without the S.O.'s prior written approval. Materials subject to welding, including backing rings to be left in place, shall be of weldable quality material. Backing ring material shall conform to the nominal chemical composition requirements of the base metal material standard.
- (c) The heat treatment for clad plate conforming to ASME SA-263 (Specification for Corrosion-Resisting Chromium-Steel Clad Plate, Sheet and Strip) and SA-264 (Specification for Corrosion-Resisting Chromium-Nickel Steel Clad Plate, Sheet and Strip) shall be done at the mill.
- (d) All clad plate, including explosion clad plate, conforming to ASME SA-263 or SA-264 shall be cold flattened, if required, only after final mill heat treatment and descaling.
- (e) Explosion clad plate not conforming to ASME SA-263 or SA-264 shall be stress relieved in accordance with the times and temperatures required by the applicable Code and the post weld heat treatment (PWHT) requirements of this specification before commencing fabrication operations. Any required cold flattening tests shall be performed before stress relieving. The use of explosion clad material requires the advance written approval of the S.O..
- (f) Pressure retaining material shall meet the tensile strength requirements of the applicable material specification after all anticipated heat treatments. Materials which require PWHT (either by Code or process requirements) shall be purchased with two simulated PWHT cycles in accordance with SA-20 Specification for General Requirements for Steel Plates for Pressure Vessels, Supplement 3, for a potential future field repair. Mill test reports shall reflect this extra cycle.

- (g) Bolting material will be specified by reference to a "S.O. Bolting Type" as listed in the following table:

Table G4.60-2: Bolting Material Specifications

S.O. Bolting Type	Bolt Material	ASME Specifications	
		Bolts	Nuts
B5	5Cr	SA193 Gr. B5	SA194 Gr. 3
B6	12Cr	SA193 Gr. B6	SA194 Gr. 6
B7	Cr-Mo	SA193 Gr. B7	SA194 Gr. 2H
B8	18Cr-8Ni Type 304	SA193 Gr. B8 Class 1	SA194 Gr. 8
B8M	18Cr-10Ni-2Mo Type 316	SA193 Gr. B8M Class 1	SA194 Gr. 8M
B8T	18Cr-8Ni Type 321	SA193 Gr. B8T Class 1	SA194 Gr. 8T
B16	Cr-Mo-V	SA193 Gr. B16	SA194 Gr. 4
CB	Carbon Steel	SA307 Gr. B	per SA307
HS	Carbon Steel	SA325	per SA325
Low Temperature Service			
B8LT	18Cr-8Ni Type 304	SA320 Gr. B8 Class 1	SA194 Gr. 8
L7	Cr-Mo	SA320 Gr. L7	SA194 Gr. 4
Sour Service			
B7M	Cr-Mo	SA193 Gr. B7M	SA194 Gr. 2HM

- (h) Gaskets shall be of materials compatible with the process fluid(s) and as specified on the vessel data sheets. The process of manufacture and dimensions or tolerances not covered by the dimensional standards shall be in accordance with the manufacturer's standards.

G4.60.6 Supplementary Materials

- (a) The manufacturer shall furnish blind flanges, gaskets, and bolting accessories for manholes, hand holes, flanged vessel ends, and for all other nozzles requiring closures.
- (b) All gaskets furnished shall be new and be installed prior to the shipment of the vessel. Gaskets, which have been used for test purposes shall be replaced with new.
- (c) One spare gasket shall be furnished for each manhole, hand hole, and flanged vessel head.
- (d) One spare set of bolting accessories of stainless steel material shall be furnished for each closure such as manholes, hand holes and flanged vessel heads.

G4.60.7 Test Requirements

(a) Code Inspection

All pressure equipment as described below shall have qualified third party inspection and certification:

(i) Vessels constructed and stamped in accordance with the ASME Boiler and Pressure Vessel Codes:

The inspector shall be qualified as required by the Code.

(ii) Vessels constructed entirely or partly to the ASME Boiler and Pressure Vessel Code without ASME Code symbol stamping:

1) The inspector shall be regularly employed for such inspection by an insurance company authorised to write boiler and pressure vessel insurance. For parts fabricated outside the United States, the inspector shall be qualified as such or be qualified by an inspection organisation recognised by the applicable governmental authority to perform pressure vessel inspection. The inspection agency selected will be subject to the S.O.'s approval.

(iii) Vessels normally within the jurisdiction of the ASME Boiler and Pressure Vessel Code constructed to standards other than the ASME Code

(iv) In all cases, the qualified inspector's certificate must be acceptable to the authority or government having jurisdiction at the plant site. The regulatory agency in the country of installation may require inspection prior to shipping.

(v) The S.O.'s inspector is not to be construed as the Authorised Inspector responsible for performing Code inspections.

(b) S.O.'s Inspection / Examination Requirements

(i) In addition to any required Code inspection, all fabrication, materials, and packaging shall be subject to inspection by the S.O. at all stages of fabrication and shall conform to the specifications.

(ii) Shop drawings of the vessel shall be available to the S.O.'s inspector at the time of the inspection.

(iii) Surfaces shall not be painted, nor the vessel shipped, until the S.O.'s inspection is complete.

(iv) Welds placed in any part (regardless of the material, thickness, or service) before the part is subjected to severe hot or cold work (ratio of thickness to local radius greater than 5 %), by any means including spinning, pressing, and rolling, shall be given a complete radiographic and either magnetic particle or liquid penetrant examination after completion of the severe working and before further fabrication is performed.

(v) The production welds attaching welding pins or studs for external insulation or fireproofing supports shall be tested by tapping with a light hammer. All such welds shall be completed prior to PWHT and/or hydrotesting. A weld procedure shall be prepared and submitted for approval prior to welding.

- (vi) Internal and external finished weld surfaces shall be either magnetic particle or liquid penetrant examined in accordance with Table 3, using the procedures of ASME Section V – Nondestructive Examination, Articles 6 and 7, respectively. Evaluations of indications shall be in accordance with ASME Section VIII – Rules for Construction of Pressure Vessels, Division 1, and Appendices 6, 7 and 8.
- (vii) Magnetic particle inspection shall not be used on non-magnetic materials. Examination shall be done after completing all required PWHT, except weld surfaces which will become inaccessible by reason of welded closures which themselves require PWHT shall be examined before such closures are welded and with prior approval of the S.O..
- (viii) When the vessel wall thickness is greater than 50 mm, the examination of accessible surfaces shall be done after the required hydrostatic test is completed.
- (ix) When liquid penetrant examination is required, it shall be performed on all accessible weld surfaces. Any surface irregularities which interfere with the examination shall be removed by grinding, machining, or with prior approval of the S.O., wash-blending with Gas Tungsten Arc Welding (GTAW).
- (x) All weld overlay, whether by manual or automatic procedures, shall be liquid penetrant examined in accordance to the methods described in ASTM E165/E165M -12.
- (xi) When the overlay involves two passes and the procedure uses an intermediate heat treatment with cooling to room temperature prior to applying the second layer, each layer shall be examined. Where overlay is to be machined, such as nozzles and flange facing, machined surfaces shall be examined. If 100 % of the overlay is examined prior to the final PWHT, the overlay shall be spot examined (no less than 20% of the surface) after heat treatment and/or after shop hydrostatic testing.
- (xii) Samples of production weld overlay shall be taken to confirm chemical analysis to the required depth. Samples shall be taken as follows to represent both manual and automatic overlay welding:
 - 1) One sample from each automatically overlaid vessel part/section (e.g., each shell ring, head, nozzle, cone, etc.); and
 - 2) Two samples from each manually overlaid vessel part/section and each manually overlaid seam (longitudinal, girth and nozzle welds).
- (xiii) A quantitative chemical analysis shall report all elements for which specific values are given in ASME Section II Materials, Part C.
- (xiv) Acceptance criteria shall be that the deposited overlay shall have the chemical composition of the applicable filler metal as specified in ASME Section II Materials, Part C. In the case of low carbon grade stainless steels, the % carbon shall not exceed 0.040 %. The Ferrite content shall be between 5-9 % (Schaeffer Diagram).

G4.60.8 Radiographic Examination

- (a) The extent of radiography shall be as indicated on the S.O.'s or approved manufacturer's vessel drawing. Radiographic examination and acceptance criteria shall be in accordance with Section VIII, Appendix 4.
- (b) Radiography shall be performed after any PWHT when PWHT is required. Radiographic film shall be fine grain, high definition, high contrast film (equivalent to Kodak type AA or better).
- (c) Film density shall be within a range of 2.0 to 3.5 as determined by film density specimens or by densitometer. Film length shall be 10 inches minimum.
- (d) Spot radiography shall be in accordance with ASME Section VIII – Rules for Construction of Pressure Vessels, Division 1, Paragraph UW-52 Spot Examination of Welded Joints, acceptance in accordance with UW-51 Radiographic Examination of Welded Joints. Spots shall be selected so that examination is made of welds representing each Welding Procedure Specification, welding machine and welding operator utilised on each vessel.
- (e) Main longitudinal, girth and head seams of vessels built of the following materials shall be 100 % radiographed:
 - (i) Carbon steel over 19 mm thick having a specified minimum ultimate tensile strength greater than 482,634 kPa (70,000 psi);
 - (ii) All chromium-molybdenum steels having over 0.5 % (nominal) chromium;
 - (iii) Austenitic steels over 19 mm thick; and
 - (iv) Carbon steel vessels with a corrosion allowance of 6.35 mm (¼ inch) and greater.

G4.60.9 Ultrasonic Examination

Ultrasonic examination shall be performed as follows:

- (a) Plate material greater than 51 mm (2 inches) but less than 100 mm in thickness shall be ultrasonically examined in accordance with ASME Boiler and Pressure Vessel Code Section V SA-435 Standard Specification for Straight-Beam Ultrasonic Examination of Steel Plates.
- (b) Plate material 101.6 mm (4 inches) thick and greater shall be ultrasonically examined in accordance with SA-435, supplementary requirement S1.
- (c) Integrally clad plate with a composite thickness greater than 12.7 mm (1/2 inch) but less than 101.6 mm (4 inches) in thickness shall be ultrasonically examined in accordance with ASME Boiler and Pressure Vessel Code Section V SA-578 Standard Specification for Straight-Beam Ultrasonic Examination of Plain and Clad Steel Plates for Special Applications, supplementary requirement S6.
- (d) Integrally clad plate with a composition thickness of 101.6 mm (4 inches) and greater shall be ultrasonically examined in accordance with SA-578, supplementary requirements S1 and S6.

- (e) Forged products (except cylindrical shell rings) 101.6 mm (4 inches) thick or greater shall be ultrasonically examined in accordance with SA-388. Forged cylindrical shell rings greater than 50.8 mm (2 inches) thick shall be ultrasonically examined in accordance with ASME Boiler and Pressure Vessel Code Section II SA-388 Specification for Ultrasonic Examination of Heavy Steel Forgings. Thickness of forgings shall be as defined in ASME Section VIII, Division 2, and Paragraph AM-200.2.
- (f) All nozzle-to-vessel welds in solid wall vessel sections (not layered construction) greater than 50.8 mm (2 inches) thick that are not radiographable shall be ultrasonically examined after PWHT.
- (g) Defects detected during ultrasonic examination exceeding the acceptance standards of the applicable Code for ultrasonic examination may be repaired only with the S.O.'s written approval. Failure to obtain S.O.'s approval prior to repair of such defects shall be cause for rejection of the material.

G4.60.10 Impact Testing

The following carbon and low alloy steel material vessel components shall be impact tested using Charpy V-notch test specimens:

- (a) All plate material 76.2 mm (3 inches) thick and over shall meet the criteria as specified in SA-20 Specification for General Requirements for Steel Plates for Pressure Vessels, (refer to Table A1.15).
- (b) All forgings 76.2 mm (3 inches) thick and over (refer to ASME Section VIII, Division 1, Fig. UCS-66.3 for thickness definition) shall meet the impact requirements specified in the applicable forging specification.
- (c) For welds in material 3 inches thick and over (production impact test plates for the weld metal and Heat Affected Zone (HAZ)), the impact test temperature shall be the lower of the minimum vessel design metal temperature (MDMT) or 0 °C (+32 °F). Impact values shall meet the requirements of ASME Section VIII, Paragraph UG-84. The test specimens shall be subjected to the same heat treatment cycles plus one for field stress relieve (time and temperature) as the completed vessel.

Table G4.60-3: Examination of Weld Surface

Type of Vessel Material		Required Surface Examinations			
		X-Ray	MT or PT		
			Level of Radiography (Note iii)	Longitudinal and Girth Welds	Nozzle Welds (Note i)
Carbon and low alloy steels with 0.5 % (max.) nominal chromium content	Not post weld heat treated	Spot	No	No	No
		Full	No	Yes	No
	Post weld heat treated	Spot	No	No	No
		Full	No	Yes	Yes
Low alloy steels over 0.5 % nominal chromium content		Full or Spot	No	Yes	Yes

Type of Vessel Material	Required Surface Examinations			
	X-Ray	MT or PT		
	Level of Radiography (Note iii)	Longitudinal and Girth Welds	Nozzle Welds (Note i)	Attachment Welds (Note ii)
Austenitic steels with DMT* less than or equal to 750° F (approx. 400 °C) and thickness of 3/4 inch or less	Spot	No	No	No
	Full	No	Yes	No
Austenitic steels with DMT* greater than 750° F (approx. 400 °C) or thickness over 3/4 inch	Full or Spot	Yes	Yes	Yes
*Design Metal Temperature				

Notes to Table G4.60-3:

- (i) Nozzle welds include the welds between nozzle and the reinforcing pad, vessel and the reinforcing pad, and the nozzle and vessel under the reinforcing pad. Welds under the reinforcing pad shall be examined before the pad is attached.
- (ii) Attachment welds include structural support, external clips, tray support, and skirt and saddle attachment welds.
- (iii) Longitudinal, girth and head seam welds.

G4.60.11 Repairs

Welds with defective areas shall have:

- (a) The defect removed. Low alloys with over 0.5 % Cr and over 38.1 mm (1.5 inches) in thickness will require S.O.'s approval of the defect removal and the repair procedures for major defects defined as longer than 2T and more than T/4 in total depth.
- (b) Confirmation of defect removal by magnetic particle or liquid penetrant examination.
- (c) Repair of welding using qualified welding procedures.
- (d) Re-post weld heat treatment if required.
- (e) The finished weld repair shall be radiographed if originally required and the surface shall be re-examined by magnetic particle or liquid penetrant examination. If the repair involves serious alterations, the approval of the S.O. shall be obtained before proceeding with the repair.

G4.60.12 Testing

- (a) All vessels shall be hydrostatically tested in accordance with the applicable Code. The test pressure shall be held for at least one hour. The test pressure shall be shown on the manufacturer's drawings.
- (b) Before the hydrotest, all internal surfaces shall be cleaned by sweeping, vacuum cleaning, or other methods so that the vessel will be free of welding slag and flux, weld rod stubs, loose scale, dirt, and debris.

- (c) Vessels shall be hydrostatically tested with potable water only; salty, brackish, or raw river water shall not be used. Vessels or parts of vessels made of Austenitic Stainless steel or Austenitic stainless clad material shall be tested with water having a maximum chloride content of 30 ppm. When the piece of equipment has components that cannot be completely drained of test water, the components shall be flushed with water containing maximum 2 ppm chlorides. As much water shall be removed as possible, and freeze precautions taken.
- (d) The manufacturer shall establish a minimum metal temperature for material where there is a risk of brittle fracture during the hydrostatic test. Normally, a minimum of 15 °C (60 °F) water and metal temperature will be required for carbon and low alloy materials.
- (e) All water shall be drained after hydrostatic testing. Any standing water in Austenitic Stainless steel or Austenitic stainless clad vessels shall be removed by blowing with air or by swabbing; heat or hot air shall not be used for drying.
- (f) Attached sleeve-type liners, including cover strips, and nozzle liners shall be tested at 173 kPa (25 psig) minimum in accordance with ASME Section V Nondestructive examination, Articles T-1031 and T-1032, by introducing dry air between the lining and the base metal. The test shall be conducted before PWHT and before pressure testing of the vessel. Any liquid which may have become trapped between the lining and the base metal shall be completely removed and all test holes drilled in the alloy liner or base metal shall be seal welded.
- (g) Each nozzle reinforcing pad or each segment thereof and similar attachments shall be tested with dry air at 173 kPa (25 psig), in accordance with ASME Section V, Articles T-1031 and T-1032. All welds inside and outside the vessel shall be inspected during the test. Test holes in pads shall be left open after the test.
- (h) The service bolting and ring joint gaskets to be furnished with the vessel may be used for testing. Any material damaged during the test, and all types of service gaskets other than ring joints used in the test, shall be replaced by the manufacturer with new material. Austenitic and B7M service bolts shall not be used for the shop test.

G4.60.13 Rejection

- (a) Completed vessels, parts of vessels, or materials containing defects originating with the manufacturer's design, materials, or workmanship, or that are not in complete compliance with the requirements of this and related specifications will be rejected.
- (b) Discovery of the conditions warranting rejection after inspection and acceptance of the vessel by the S.O. does not relieve the manufacturer of his responsibility to comply with the requirements of all applicable specifications.

G4.60.14 Materials and Workmanship

- (a) Equipment Specifications
 - (i) Vessels shall be furnished as specified herein;
 - (ii) General arrangement of the vessels, including locations and orientations of nozzles, piping, access platforms, stairs, ladders, accessories, and other appurtenances shall be as indicated on the drawings; and

- (iii) The numbers, locations, types, and sizes of the piping connections indicated on the drawings are indicative only and subject to change later by the S.O.. or confirmation by the Contractor depending on the design scope ownership.
- (b) Code Requirements
 - (i) Vessels shall be designed and constructed in accordance with the latest editions of the specified codes or standards. ASME code stamping shall be applied where specified; and
 - (ii) Data reports for all code stamped tanks shall be provided. ASME code stamped tanks shall be registered with the National Board of Boiler and Pressure Vessel Inspectors.
- (c) Code Stamping and Nameplates
 - (i) All vessels shall be furnished with a stainless steel nameplate. Required markings shall not be stamped directly on the vessel;
 - (ii) Nameplates shall be installed on the manufacturer's standard nameplate holder of sufficient length to project at least 1 inch beyond the vessel insulation;
 - (iii) The manufacturer shall assign and stamp a National Board Registration number on all vessels manufactured and stamped to ASME Section VIII;
 - (iv) The S.O.'s equipment item number shall be stamped on the nameplate;
 - (v) The nameplates of vertical vessels shall be located on the shell above the lowest manhole; and
 - (vi) On horizontal vessels, they shall be located on the side at the centreline or above a manhole. The nameplate location shall be shown on the manufacturer's drawings.
- (d) Construction
 - (i) All vessels shall be of the all-welded construction and all shall have lifting lugs; and
 - (ii) All welds, welding, and related operations shall be in accordance with the corresponding specifications described herein.
- (e) Shell and Head Seams
 - (i) All vessel seams shall be located to clear longitudinal support welds for internals;
 - (ii) Longitudinal seams in cylindrical and conical shells and all seams in spherical shells and built-up heads shall be located to clear openings and reinforcing pads.
 - (iii) Where possible, circumferential seams in cylindrical and conical shells shall be located to clear openings and reinforcing pads. Weld seams that will be covered by reinforcing pads shall be ground flush and 100 % radiographically examined to 6 inches beyond the edge of the pad before attaching the reinforcing pad.
 - (iv) Joints in horizontal vessels shall not be located coincident with or across saddle supports.

- (v) Where possible, structural attachment welds shall clear weld seams by at least one plate thickness. If overlap is unavoidable, the portion of the seam to be covered shall be ground flush and 100 % radiographically examined to 152.4 mm (6 inches) beyond the edge of the weld before the attachment is welded on.

(f) Reinforcement of Openings

The following reinforcing requirements shall apply (except as noted below) to vertical vessels having H/D ratios greater than 5, where H is the vessel length between tangent lines and D is the vessel inside diameter (or the smallest diameter of vessels having more than one diameter). These requirements do not apply to NPS (Nominal Pipe Size) openings 3 inches and smaller that are not required to have reinforcement by ASME Section VIII, Division 1, Paragraph UG-36(c) (3):

- (i) The total cross-sectional area of reinforcement required shall not be less than:

$A = d \times t$ (in²), where:

$d =$ the diameter of the finished opening in its corroded condition (in).

$t =$ the nominal thickness of the shell less corrosion allowance (in).

All metal area within the limits of reinforcement, except the shell metal area, may be considered as having reinforcing value as permitted by the Code, unless otherwise specified on the drawings.

- (ii) The strength (allowable stress) of the attachment welds joining the vessel wall and the reinforcement or any two parts of the attached reinforcement shall be at least equal to the smaller of:

- 1) The strength in tension of the cross section of the area of reinforcement, or

- 2) The strength in tension of area A as defined above.

- (iii) Each reinforcing pad, or segment, if more than one piece is used, shall be provided with two ¼ inch NPT (Nominal Pipe Thread) threaded holes. Welds in segmented two-piece reinforcing pads on cylinders and cones shall be in the vessel's circumferential direction. The outside edge of reinforcing pads shall be ground or otherwise made reasonably smooth before welding to the vessel.

(g) Nozzles and Connections

- (i) Flange bolt hole orientation shall straddle normal centrelines.

- (ii) All nozzles, except nozzles with attached internal piping, liquid outlet nozzles (without internal standpipes), vents, vapour outlets and drain nozzles, may have a 12.7 mm (1/2 inch) maximum internal projection. Liquid outlet nozzles, drain nozzles, and manholes shall be set flush with the inside of the vessel. The inside corner of nozzle necks shall be rounded to 3.175 mm (1/8 inch) radius, minimum.

- (iii) Raised and flat face flanges shall be furnished with a stock gasket surface finish unless otherwise specified on vessel drawings or data sheets.

- (iv) When tongue and groove facing is used, the groove shall be in the nozzle flange; when the nozzle is located in the bottom head of a vessel, the groove shall be in the cover.
- (v) The inside diameter of nozzles shall be no smaller than the inside diameter of double extra strong pipe for NPS 25 mm to NPS 75 mm nozzles inclusive, Schedule 160 pipe for NPS 100 to NPS 300 mm nozzles inclusive, and nominal diameter minus 50 mm for nozzles over NPS 300 mm.
- (vi) In general, flanges shall be raised face drilled to BS EN 1092 according to the required pressure rating. With prior approval of the S.O., special flanges may be designed in accordance with ASME Section VIII, Division 1, and Appendix 2.
- (vii) Slip-on flanges shall not be used for flange ratings above PN10. When slip-on flanges are permitted under special conditions and with S.O. written approval, the ring between the slip-on flange and the nozzle neck shall be vented by a 3 mm diameter vent hole. Lap joint flanges shall not be used without written approval from the S.O..
- (viii) Internal flanges for non-pressure connections may be of plate construction with machined flat faces. Gaskets are not required, except when specified otherwise.
- (ix) External threaded connections shall be 275 kg full couplings. External threaded connections are limited to NPS 13 mm maximum size. Couplings are not allowed on vessels with a corrosion allowance larger than 3 mm.
- (x) All threaded connections shall have NPT threads conforming to BS EN 10226 Parts 1 to 3 or ASME B1.20.1, formerly ANSI B2.1.
- (xi) When full radiography is specified on the vessel datasheets, this shall be interpreted as 100 % radiography (RT-1), not RT-2.
- (xii) In sulfide stress corrosion (SSC) services, "cold forming" of fittings and end closures is prohibited.
- (xiii) Intermediate heads as shown in Fig.UW-13.(1).(f) of the Code shall not be used, if the shell thickness of the vessel exceeds 15.88 mm (5/8 inch).
- (h) Manholes, Handholes and Covers
 - (i) Manhole, handhole, and flanged vessel end covers or blind flanges shall be in accordance with BS EN 1092 or, with prior approval of the S.O., may be designed in accordance with Code rules for the specified design and test conditions of the vessel. The final flange thickness shall include the specified corrosion allowance. BS EN 1092, ANSI or API blind flanges may be used on vessels with a corrosion allowance of 6 mm or less without further verification.
 - (ii) All covers and blind flanges weighing over 25 kg shall be furnished with davits or hinges.
 - (iii) The inside edge of the manway bore shall be rounded to a minimum radius of 3 mm.
 - (iv) Manhole openings in a horizontal plane shall be furnished with a handgrip inside the vessel above the manhole.

- (i) Corrosion Allowance/Protection
 - (i) The corrosion allowance for pressure boundary parts shall be as specified on the vessel drawing.
 - (ii) The minimum shell and head thickness, exclusive of corrosion allowance, shall be 4.76 mm for carbon and low alloy steel vessels and 3.175 mm for high alloy steel vessels. Minimum composite thickness of clad shell or head materials shall be 9.53 mm.
 - (iii) The minimum thickness of a skirt, saddle web or angle leg is 6.35 mm exclusive of corrosion allowance.
 - (iv) In alloy, alloy lined or clad vessels (or sections), all surfaces of pressure parts and fixed internals shall either be fabricated of or protected by the alloy or alloy lining material specified for the vessel.
 - (v) In unlined vessels (or sections), all surfaces of pressure parts and fixed internals shall have the specified vessel corrosion allowance added to each surface.
 - (vi) The minimum corroded thickness of fixed internals shall be 6.35 mm, and the thickness of fully corroded fillet welds measured through the throat shall be 1.25 mm as minimum. For removable internals, the specified corrosion allowance shall be reduced by half, and the minimum corroded thickness shall be 3 mm as minimum.
- (j) Vessel Internals
 - (i) Removable internal parts shall be designed in units as large as can be installed and removed from the vessel through the nearest manhole opening indicated for that purpose. When internals are obviously not able to be removed through a manhole (e.g., NPS 600 mm internal piping in a vessel with an NPS 600 mm manhole), they shall have corrosion allowance added in accordance with General specification on Painting and Protection.
 - (ii) The tray manufacturer shall fabricate trays and removable tray supports required as internals for vessels and, unless otherwise noted on the vessel drawings or data sheets, shall be shop installed by the vessel manufacturer.
 - (iii) All welded-in tray supports shall be designed, detailed and fabricated by the tray manufacturer, but installed by the vessel manufacturer.
 - (iv) Tray and tray support installation shall be in accordance with the tray manufacturer's drawings and recommendations.
 - (v) The installer shall exercise sufficient care and make inspections during bolting and packing to provide assurance that leakage tests (bubble cap trays) or levelling required by the applicable standard will be met in the field:
 - 1) The S.O. shall be notified one week before tray installation is started;
 - 2) Trays may be installed at any convenient period during the fabrication operations after post weld heat treatment is completed;
 - 3) Tray manways shall be installed after the S.O. has completed all inspections. All gaskets shall be installed and bolting fully tightened before shipment;

- 4) Internal non-pressure carbon and low alloy (6 % Cr maximum) steel pipe and fittings removable through the vessel manhole shall have the following minimum nominal wall thickness or ratings:
 - Pipe NPS 304 mm (12 inches) and smaller: standard wall thickness; and
 - Pipe NPS 356 mm (14 inches) and larger: 1/4 inch wall thickness.
 - 5) Internal non-pressure medium-alloy and high-alloy (8 % Cr and over) steel pipe and fittings shall have the following minimum nominal wall thickness or ratings:
 - Pipe NPS 152 mm (6 inches) and smaller: Schedule 40S thickness; and
 - Pipe NPS 204 mm (8 inches) and larger: Schedule 10S thickness.
- (k) Linings
- (i) Vessels requiring a lining shall be either weld overlaid or integrally clad. Strip-type linings shall not be used unless specified by the S.O.. Use of an Austenitic lining on carbon or low alloy vessels above 480 °C will require S.O.'s review and approval due to loss of corrosion resistance due to carbon migration.
 - (ii) When permitted, strip lining shall be attached to the shell or nozzle circumferentially and shall be designed so that the width of the strips does not exceed that shown in Table G4.60-4. The welds between the strips shall be at least 6.36 mm (1/4 inch) and not more than 12.7 mm (1/2 inch) in width.

Table G4.60-4 - Strip Welded Linings

Vessel Design Temperature	Width of Strips
343.3 °C (650 °F) and under	101.6 mm – 12.7 mm (4-1/2 inches)
Over 343.3 °C (650 °F) to 426.6 °C (800 °F)	101.6 mm (4 inches)

- (iii) Where vessel temperatures exceed 425 °C (797 °F), the width of strips will be shown on the design drawings.
 - (iv) Vessels with integrally clad linings shall be made of plate clad with the required lining material at the steel mill in accordance with the applicable material specifications.
 - (v) Explosion clad material shall not be used without the written consent of the S.O..
- (l) Supports
- (i) Vessel supports shall follow dimensioned design sheets which may be available at a later timing as the specifications. If not, the Contractor and Supplier shall be responsible for the detail design and verify the adequacy of the supports for the design conditions specified.
 - (ii) Sufficient support lugs shall be provided.
 - (iii) Skirt supported vessels shall have all skirt access openings reinforced and pipe penetrations shall be sleeved.

- (iv) Vertical vessels supported by skirts shall conform to the following requirements:
 - 1) The centreline of the skirt and the corroded shell plate (bottom course) shall be approximately coincident, except for conical skirts attached to the shells of vertical vessels;
 - 2) When necessary to maintain the calculated stresses within the allowable stresses, skirts may be attached to the flanged portion of the head. The skirts shall also be positioned so that the vessel shell-to-head girth seam can be inspected;
 - 3) Vessels requiring field hydrotest shall have the skirts designed for such tests; and
 - 4) Skirts of vessels requiring field stress relieve shall be verified for conditions and loading during stress relief and shall require S.O.'s approval.
- (m) Lifting Attachments
 - (i) Two lifting lugs shall be provided for vertical vessels, unless stated otherwise on the vessel drawing.
 - (ii) Each lug shall be designed for 1.5 times the erection weight of the vessel, utilizing an impact factor of 1.5. The actual dead load reaction at initial pick shall be calculated based on the vessel's centre of gravity. The design shall account for all bending moments and shear forces as the vessel is lifted from the horizontal to the vertical.
- (n) Fabrication
 - (i) Fabrication shall not begin until the following two milestones have been met:
 - 1) Fabricator has received drawings and weld procedures from S.O. that have been APPROVED or APPROVED AS NOTED; and
 - 2) A Prefabrication Meeting has been conducted by the S.O.'s Inspection Department.
- (o) Design of Welded Joints
 - (i) Pressure-retaining shell, head, and transition section seams, nozzle and manway attachments to shell, head, or transition section and similar joints shall be full penetration double-welded joints except as noted above. (Nozzles shall be inserted through the wall of the vessel; nozzles abutting the vessel wall (set-on) shall not be used.) This requirement is not intended to prohibit reinforcing pad attachment details similar to ASME Section VIII, Division 1, and Figure UW-16.1 (h), provided the requirements of UW-41 are met. Other joints are subject to approval by the S.O..
 - (ii) When sound back-welding is not feasible, single welded joints with backing strips or rings shall be used; nozzle and manhole attachments in plates 9.53 mm (3/8 inch) thick and less may be single welded joints, using a TIG root pass without backing. Backing strips or rings shall not be used when corrosion allowance is 3.18 mm (1/8 inch) or more. Backing rings shall not be used without prior written approval of the S.O..

- (iii) Joints in alloy steels shall be made without the use of permanent backing strips. Temporary backing strips may be utilized if, after removal, the joint is examined visually for contour and undercutting and for cracks by the magnetic particle or the liquid penetrant method of examination. These examinations shall be performed after post weld heat treatment (PWHT), if PWHT is required.
- (p) Preparation for Welding
 - (i) On alloy clad plates, if flame cutting is used for preparing plate edges, any flame-gouged areas at the junction of the alloy and the base metal shall be repaired by stripping the cladding back at least 9.53 mm (3/8 inch) from the gouged area, and building up the base metal with weld deposits, and then the clad surface with alloy weld metal.
 - (ii) Weld metal shall not be used to build up the edges of plates that are too short or those that contain large cavities or weld root gaps over 3.175 mm (1/8 inch), without advance written approval of the S.O..
 - (iii) The maximum allowable gap between nozzle and manhole reinforcing pads and the curvature of the vessel shells or heads before welding shall be 3.175 mm (1/8 inch).
 - (iv) The weld bevels of stub-end nozzles shall be in accordance with ASME Section VIII.
 - (v) All weld bevels in material 12.7 mm (1/2 inch) thick and over shall be examined by the Magnetic Particle or Liquid Penetrant method. Any laminations exceeding the limits specified in Paragraph 9.3.5 of ASME SA-20 will require the S.O.'s advance written approval.
 - (vi) All nozzle cut outs in shell and head materials with a wall thickness over 50.8 mm (2 inches) thick shall be ultrasonic test (UT) inspected for lamination prior to cutting out the nozzles. Any lamination exceeding the limits specified in Paragraph 9.3.5 of ASME SA-20 20 Specification for General Requirements for Steel Plates for Pressure Vessels shall require S.O.'s advance written approval before cutting.
- (q) Welding
 - (i) Welding shall be in accordance with ASME Section IX Welding and Brazing Qualifications.
 - (ii) Fabrication involving welding shall not be sublet to others without advance written approval of the S.O..
 - (iii) Temporary attachment welds on pressure shells shall be removed. The surface under such welds, and under backing rings which have been removed shall be properly ground or conditioned to eliminate surface stress raisers. Such surfaces shall be examined by either the Magnetic Particle or Liquid Penetrant method of examination when the surface material is carbon steel with a design metal temperature below 0 °C (32 °F); Austenitic alloy steel with a design metal temperature above 400 °C(750 °F); or Ferritic alloy steel. When the surface material is carbon steel with a design metal temperature of 0 °C (32 °F) and above, such surfaces shall be so examined if attachment weld surfaces are required to be examined in accordance with Table 3.

- (iv) When arc strikes occur, the surface shall be properly ground or conditioned to eliminate surface stress raisers. These surfaces shall be examined by either the Magnetic Particle or Liquid Penetrant method of examination. Any defects found shall be removed and the surface repaired and re-examined.
- (r) Brinell hardness
 - (i) The Brinell Hardness of the weld metal and parent metal of the heat affected zone (HAZ) taken after any required PWHT shall not exceed the limits set in Table G4.60-5 (one (1) test per longitudinal or girth seam).

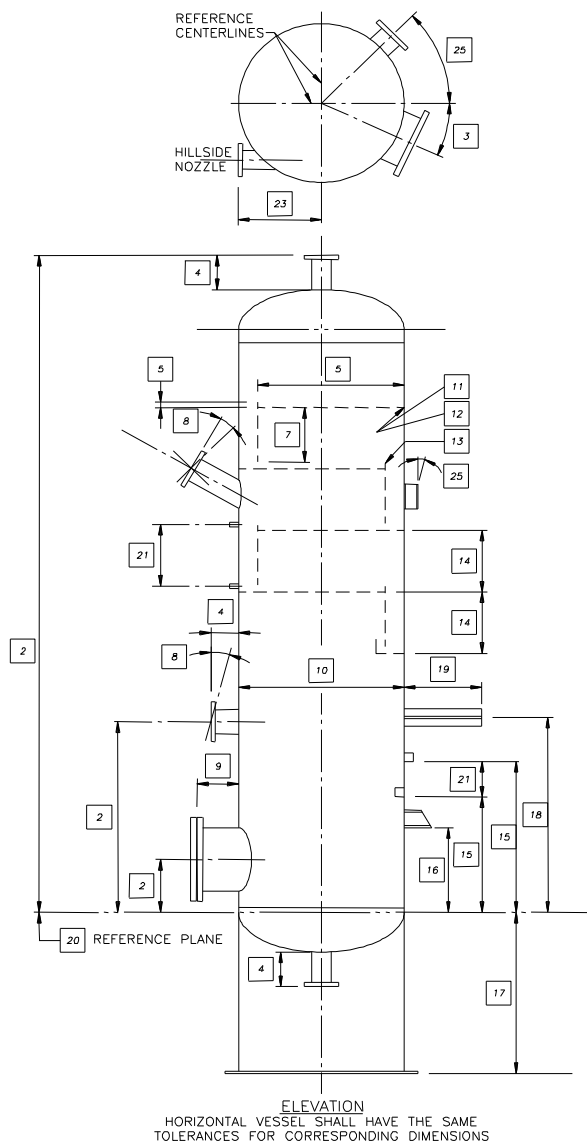
Table G4.60-5

Material P-Number	Maximum Brinell Hardness	Average of all Readings of Individual Welds
P1 Groups 1, 2, 3	215	200
P1 Group 4	250	230
P3, P4	225	210
P5	235	220
P6, P7	240	240
P10A, P10B	225	210

- (s) Weld overlay/clad restoration/strip liner attachment.
 - (i) Weld overlay shall be applied circumferentially and shall have a relatively smooth surface without notches, flaws, or undercuts that would act as stress raisers. Small bore nozzles may be overlay welded longitudinally followed by grinding or machining to a smooth surface.
- (t) Post weld heat treatment
 - (i) Post weld heat treatment (PWHT) shall be performed as required by the Code or when specified in the data sheets. PWHT shall be performed in a furnace. Internal gas firing or external electrical wrapping is not allowed unless specifically approved by the S.O.. The entire vessel shall be given PWHT in one heat. Gasket seating surfaces of flanges shall be protected against oxidation and scale formation during PWHT.
 - (ii) A PWHT procedure including number and location of thermocouples shall be submitted to the S.O. for review. Maximum temperature differential between any two thermocouples shall not exceed 38 °C (100 °F)
 - (iii) Post weld heat treatment for welds between dissimilar metals shall conform to the Code requirements and shall be verified by a Procedure Qualification Test. The proposed post weld heat treatment for welds between Ferritic and Austenitic steels shall be reviewed by the S.O..
 - (iv) Local PWHT shall not be performed without advance written approval of the S.O..

- (v) P-3 (C-Mo and Mn-Mo) materials shall be post weld heat treated at a temperature of 593 °C (1100 °F) minimum. Note 1 to Tables PW-39 of ASME Section I, UCS-56 of ASME Section VIII, Division 1, and AF-402.1 of ASME Section VIII, Division 2 shall not apply to P-3 (C-Mo and Mn-Mo) materials.
- (vi) Carbon steel vessels in wet/sour service (H₂S with H₂O) and of P1 material shall be PWHT at 620 °C (1150 °F) minimum, and shall follow Specification Q511, Additional Material and Fabrication Requirements for Carbon Steel Vessels in H₂S/Amine Service.
- (vii) Plate, seamless heads, parts of built-up heads, and similar pressure-retaining parts subjected to cold or hot bending and forming shall be heat treated as follows:
 - 1) As required by the applicable Code and the applicable material specification;
 - 2) The hot forming temperature may be used as the normalizing temperature if the forming temperature is verified to be at or above the normalizing temperature at the completion of forming.
 - 3) Where the Code or specification requires normalizing, the hot-formed parts which do not meet the requirements of sub-paragraph 2 above shall be normalized after forming.
- (u) Tolerances
 - (i) Unless otherwise noted on the drawings, tolerances shall conform to the requirements of the applicable Code or the following figure, whichever is more stringent:

Figure 1: Tolerances



Notes for Figure 1:

- (ii) The tolerance on dimensions not specified herein or specifically shown on the drawings shall be ± 6.35 mm (1/4 inch).
- (iii) Items not covered by Notes (xvi), (xvii), and (xviii) which have been located on the S.O.'s drawings by elevation or by a dimension from a tangent or reference line shall be located from the reference plane, within a tolerance ± 6.35 mm (1/4 inch).
- (iv) The tolerance on orientation from the reference centreline to the centreline of a nozzle or manhole is plus or minus 1 degree, except that the circumferential measurement shall have a maximum tolerance of ± 25.4 mm (1 inch).

- (v) The tolerance on the distance from the face of a nozzle to the vessel surface is ± 6.35 mm (1/4 inch).
- (vi) The tolerance on the distance from the inside vessel wall to downcomer support bars or weirs is as shown on the tray manufacturer's tower attachment drawing.
- (vii) The tolerance on the height of downcomer bolting bars above tray support rings is as shown on the tray manufacturer's tower attachment drawing.
- (viii) The tolerance on the length of downcomer bolting bars is as shown on the tray manufacturer's tower attachment drawing.
- (ix) The tolerance on the alignment of a nozzle flange face with the indicated plane is ± 0.5 degree in any direction.
- (x) The tolerance on the distance from the face of a manhole flange to the vessel surface is ± 12.7 mm (1/2 inch).
- (xi) The maximum deviation of the actual circumference (as determined by strapping) from the nominal circumference is as shown in Column A, Table G4.60-6.
- (xii) The tray support rings shall be perpendicular to the longitudinal axis of the vessel within the limits given in Column B, Table G4.60-6.
- (xiii) The top surface of tray support rings shall be perpendicular to the vessel shell within 1.58 mm (1/16 inch) in the width of the ring.
- (xiv) The top of weld-attached weir plates shall be perpendicular to the longitudinal axis of the vessel within the limits given in Column C, Table G4.60-6.
- (xv) The tolerance on the distance between adjacent tray supports is ± 3.17 mm (1/8 inch).
- (xvi) The maximum deviation in elevation between each adjoining clip, bracket, and similar structural attachment located in the same plane is ± 3.17 mm (1/8 inch). The tolerance on location from the tangent or reference line for each individual clip, bracket, or attachment is shown in Note (ii).
- (xvii) The tolerance on the distance from the reference plane to support lug is ± 6.35 mm (1/4 inch).
- (xviii) The tolerance on the distance from the reference plane to the base is ± 6.35 mm (1/4 inch).
- (xix) The tolerance on the distance from the reference plane to horizontal vessel support (saddle) is ± 6.35 mm (1/4 inch).
- (xx) The tolerance on the distance from the vessel surface to the bottom of horizontal vessel support saddles is ± 6.35 mm (1/4 inch).
- (xxi) The reference plane shall be located a minimum of 50.8 mm (2 inches) above the root land in a vertical vessel bottom course and 50.8 mm (2 inches) from the root land in a horizontal vessel end course at the side from which dimensions are run. The plane shall be punch marked inside and outside at 0, 90, 180, and 270 degree points. Punch marks shall be circled with paint.

- (xxii) The tolerance on the specified dimension between two nozzles, connections, or clips is ± 3.17 mm (1/8 inch).
- (xxiii) The maximum deviation from a straight line perpendicular to the reference plane of a cylindrical element of the vessel shell (including cylindrical or similar supports extending beyond the shell of vertical vessels) shall be 3.17 mm (1/8 inch) for each 3 m (10 feet) of length.
- (xxiv) The tolerance on the distance from the face of a hillside shell nozzle to the vessel centreline is ± 6.35 mm (1/4 inch).
- (xxv) The tolerance on nozzle bolt hole orientation is ± 1.58 mm (1/16 inch) measured at the bolt circle diameter.
- (xxvi) The tolerance for external clip orientation and clip face alignment is as shown in Notes iii and viii.

Table G4.60-6 - Variable Tolerances

Column/Tower, Diameter,	Tolerances,		
	A	B	C
1.22 m (4 ft) and less	9.52 mm (3/8 inch)	3.17 mm (1/8 inch)	3.17 mm (1/8 inch)
Over 1.22 m (4 ft) to 2.43 m (8 ft)	12.7 mm (1/2 inch)	6.35 mm (1/4 inch)	4.76 mm (3/16 inch)
Over 2.43 m (8 ft) to 4.57 m (15 ft)	25.4 mm (1 inch)	7.93 mm (5/16 inch)	6.35 mm (1/4 inch)
Over 4.57 m (15 ft)	12.7 mm (1/2 inch)	9.52 mm (3/8 inch)	7.93 mm (5/16 inch)

- (xxvii) Permissible out-of-roundness of cylindrical, conical, and spherical shells for internal and external pressure will be inspected in accordance with Paragraph UG-80 of the ASME Section VIII, Division 1.
- (xxviii) The tolerances on the specified thickness of the alloy on integrally clad plate shall be as given in Table G4.60-7.

Table G4.60-7 - Cladding Tolerances

Sections	Tolerance for the Clad Thickness	
	Nominal Thickness	Minimum Thickness
Flat plate as rolled, cylindrical & conical sections after rolling and other formed sections	Under Tolerance - 12.5 %	Under Tolerance - 0 %
	Over Tolerance (By Supplier)	Over Tolerance (By Supplier)

G4.60.15 Painting, Marking, Packaging, and Shipping

- (a) Painting.
 - (i) Vessels shall be shop painted in accordance with the requirements of the vessel drawing or data sheets.
 - (ii) Austenitic alloy steel and nickel-iron-chromium alloy material parts with design metal temperatures above 400 °C (750 °F), except as required by section G4.60.17 Protection during Shipment, shall be kept free of paint to avoid the possibility of corrosion at elevated operating temperatures. All traces of paint which accidentally spatter on such parts shall be promptly removed.

G4.60.16 Marking

- (a) Stamping on all Austenitic steels and impact tested materials shall be with "low-stress" steel stamps having round or "U" shaped cross sections or with "interrupted-dot" die stamps.
- (b) Marking paint or water insoluble ink for use on Austenitic and high nickel alloy steels shall contain no substance (e.g., metallic pigments, sulfur or chlorides) which could be harmful to these materials at ambient or elevated temperatures. The manufacturer shall submit analysis of marking material to the S.O.. Submittal shall demonstrate (by chemical analysis and history of use) that the material meets these requirements.
- (c) Internals, clips, lugs, or other similar attachments which are removable and are to be assembled by others shall be marked with piece numbers for identification in assembly and match-marked.
- (d) For vessels that have been PWHT, the **"NO WELDING PERMITTED ON THIS VESSEL"** sign shall be painted on the vessel after heat treatment with lettering size not less than 77 mm (3 inches) height using white paint.
- (e) On vertical vessels, this sign shall be located on two opposite sides near the bottom tangent line and repeated at approximately each 3 m (10 feet) of height, but rotated 90 degrees.
- (f) On horizontal vessels, the sign shall be located on both sides near the horizontal centreline.
- (g) The North and East orientations shall be centre-punched on the outside of each vertical vessel approximately 152.4 mm (6 inches) above the bottom tangent line and 152.4 mm (6 inches) below the top tangent line. The punch marks shall be circled with white paint.

G4.60.17 Protection During Shipment

- (a) The vessel shall be free of debris, dirt, and foreign matter, and be thoroughly cleaned inside and out before shipment.
- (b) The manufacturer shall be responsible for suitably packaging each vessel or component and adequately supporting and securing all vessel internals to protect from damage or loss during handling and shipment in accordance with the Purchase Order and following requirements:
 - (i) All materials and methods of packing shall conform to the requirements and specifications described or referenced in the Purchase Order;
 - (ii) Packages shall include lifting lugs or designated lifting points;
 - (iii) Centres of gravity shall be clearly shown;
 - (iv) Each vessel, crate, or bag shall be durably marked with the receiving address, Purchaser's vessel Item Number, and the complete Purchase Order Number. All markings shall be in the English language;
 - (v) Special tools shall be separately boxed, shipped and identified with the vessel on which they are to be used, and the boxes marked as special tools;

- (vi) The manufacturer shall include a packing list with each shipment, listing the contents by assembly piece mark, individual piece mark, or by vessel number of each box, crate, bag, and skid. The list shall state whether the contents constitute complete or partial shipment;
- (vii) Vessels shall not be transported to the jobsite as above-deck cargo in salt water conditions without prior written approval of the Contractor and S.O.;
- (viii) Where clearances for shipping require alteration or cutting off of vessel nozzles, clips, lugs, or other similar attachments, the manufacturer shall submit loading diagrams showing the attachments which require re-orientation or removal. No changes shall be made unless written approval has been obtained from the S.O.;
- (ix) Small parts of the vessel shall be shipped bagged, boxed, or otherwise protected from damage or loss. Spare gaskets shall be shipped separately and not bolted in place. They shall be clearly tagged as spares;
- (x) The manufacturer shall provide any special protection or packaging, and details of any storage, shelf life, or maintenance instructions which are not within the scope of the Purchase Order, but which pertain to the manufacturer's guarantee or are otherwise necessary for protection of the vessel;
- (xi) Protection for flange faces, thread connections, weld bevels, etc., shall be as follows:
 - 1) All machined and threaded surfaces of carbon steel and Ferritic alloy steel material, except weld bevels, shall be coated with a suitable rust inhibitive coating approved by the S.O.; and
 - 2) Weld bevels shall be free of dirt, oil, grease, scale, rust, and other foreign materials. All weld bevels of carbon steel and Ferritic alloy steel materials shall be coated after cleaning, on the inside and outside for a distance of approximately 77 mm (3 inches) from the end of the weld bevel with a suitable weldable rust inhibitive coating approved by the S.O..
- (xii) All weld bevels shall be closed with metal or plastic caps to prevent damage and entrance of foreign materials;
- (xiii) All socket-weld ends and plain ends shall be closed with metal or plastic protectors that fit either inside or outside to prevent damage and entrance of foreign materials;
- (xiv) Threaded openings shall be plugged with hexagonal headed threaded plugs of the same material as the connected part and sealed with Tetrafluoroethylene (TFE) tape thread sealant;
- (xv) Flanged openings shall be protected and made water-proof with suitable plastic flange covers or full size steel covers and rubber gaskets between the flange face and cover; and

- (xvi) When steel flange covers are furnished, they shall be secured with appropriately sized machine bolts as follows:
- 1) For flanges having 4 to 28 bolt holes, a bolt shall be placed in at least every other hole, with a minimum of four bolts; and
 - 2) For flanges having more than 28 bolt holes, bolts shall be placed in at least every fourth bolt hole.

END OF SECTION